INCH POUNDS

A-A-59141C
July 7, 2022
SUPERSEDING
A-A-59141B
July 7, 2017

COMMERCIAL ITEM DESCRIPTION

TRUCK, LIFT, FORK, ELECTRIC, PNEUMATIC-TIRES, 2000 POUNDS CAPACITY AT 24 INCH LOAD CENTER, 144 INCH LIFT HEIGHT

The General Services Administration has authorized the use of this commercial item description as a replacement for type "EE" of MIL-T-21869-35 for all federal agencies.

1. SCOPE. This commercial item description (CID) covers the general requirements for an electric, front wheel drive, rear wheel steer, pneumatic-tires, 2,000 pound at 24 inch load-center capacity, forklift truck. The forklift is intended for handling cargo in and around warehouses, loading platforms, and docks; on paved or other hard surfaces.

2. CLASSIFICATION. The forklift shall be type E, unless otherwise specified as type EE or type EX, in accordance with Underwriters Laboratories (UL) 583.

3. SALIENT CHARACTERISTICS

3.1 Safety. The forklift shall comply with ANSI/ITSDF B56.1 and OSHA standards in effect at the time of manufacture. A seat belt conforming to SAE J386 and restraint devices designed to ensure the operator's upper body remains entirely within the protection of the overhead guard in the event of tip-over shall be provided. Unless otherwise specified (see 7.3), the final color shall be yellow for safe operation in an indoor environment. The forklift shall be type E, unless otherwise specified as type EE or type EX, in accordance with UL 583. A mounted fire extinguisher shall be provided.

3.2 Environmental requirements. The forklift shall operate as specified herein in any ambient temperature from 0 to 120 °F.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data that may improve this document should be sent to: AFLCMC/EZSS, Bldg. 28, 2145 Monahan Way, Wright-Patterson AFB, OH 45433-7017 or email engineering.standards@us.af.mil. Since contact information can change, you should verify the currency of this information using the ASSIST Online database at https://assist.dla.mil.

FSC 3930

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3.3 Battery. The battery shall have adequate capacity to provide not less than 7 hours of continuous operation under normal, industrial usage. The battery shall have “SB” type half connectors. When specified (see 7.3), the forklift shall be shipped with a battery and/or battery charger.

3.4 Battery discharge indicator. A battery discharge indicator shall be located so that it is visible to the seated operator.

3.5 Controls. All controls shall be in accordance with ANSI/ITSDF B56.1. All load motion controls shall be self-centering, that is they shall return to the neutral position when released.

3.6 Hydraulic system. A pressure relief protection device is required along with pump(s), cylinders, control valves, filter(s), reservoir, hoses, and all other components necessary to make a complete hydraulic system.

3.7 Uprights and carriage. With no load on the forks, the seated operator shall be able to see at least one fork tip at all lift heights.

3.8 Forks and carriage. The forks shall be mounted so that when fully lowered and with the forklift operating in reverse, there is no damage or degradation to the fork tines, there attachment, or any other component of the carriage, including fork positioner and side shift.

3.8.1 Standard forks. The fork tine dimensions shall be:

a. Length: 36.0 ±1 inches
b. Width: 4.06 inches
c. Thickness: 1.56 inches.
d. Tip thickness: Not more than 0.5 inches
e. Underside of forks taper: Not less than 14 inches

3.8.2 Fork positioner. The forks shall be manually adjustable with spacing of 14 inches (or less) when in the most closed position, and 30 inches (or more) when in the most spread position. These dimensions shall be measured to the outside edges of the forks.

3.8.3 Side shift. Hydraulically powered side shifting controlled by a seated operator, with at least 4 inches of side shift on each side of the center, (8 inches minimum total travel) and with a rated load on the forks shall be furnished.

3.9 Load backrest. The load backrest, in conjunction with the forks and carriage, shall provide a vertical rear load guard of at least 48 inches high, measured from the load horizontal carrying surfaces of the forks.

3.10 Steering. Power steering shall be furnished.

3.11 Service brakes. Service brakes shall conform to ANSI/ITSDF B56.1.

3.12 Parking brake. Parking brake shall conform to ANSI/ITSDF B56.1.

3.13 Electrical system. The electrical system shall consist of all components necessary for the operation of the forklift.

3.14 Instruments. The instruments, gauges or warning indicator lights that are on the manufacturer’s standard commercial model shall be furnished. In addition, an unlighted hour meter capable of totaling at least 9999 hours shall be installed in the instrument panel.

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3.15 External lights. A minimum of two floodlights shall be furnished. One in front, facing forward, and one mounted in the rear, facing rearward. One combination tail light and brake stop light shall be installed. All lights shall be protected by location placement or guards. Individual operator controlled switches shall be provided for front and rear floodlights.

3.16 Horn. A horn shall be furnished with the horn actuator located in the center of the steering wheel.

3.17 Back-up alarm. An audible back-up alarm shall be installed on the forklift. The back-up alarm shall automatically activate when the transmission selector is placed in reverse. The back-up alarm must conform to SAE J994.

3.18 Operator's overhead guard. The guard shall be in accordance with ANSI/ITSDF B56.1, including inside clearances. It shall also be a falling-object-protective-structure (FOPS) conforming to ANSI/ITSDF B56.1. The height, from the ground to the top of the overhead guard shall be not more than 86 inches.

3.19 Seat. The forklift shall have a vinyl, mechanical-suspension seat with an operator presence system.

3.20 Painting. The forklift shall be primed and painted using standard commercial practices. The final paint shall be a standard, commercially-available polyurethane. The paint finish shall show no evidence of uneven application, curtains, runs, snags, orange peel, lack of adhesion, or other defects. Unless otherwise specified (see 7.3), the final color shall be yellow.

3.21 Walkway coating. Floor plates and step surfaces shall be made of, or covered with, a non-slip metal or tread-plate surface.

3.22 Slinging and tie-down provisions. There shall be permanently installed slinging devices that enable the forklift to be lifted in its normal travel configuration, and tie-downs that enable the forklift to be tied down to the floor of the transportation media. All devices shall be capable of accepting a chain and hook with an outside diameter of 3 inches and shall have a structural safety factor of at least 2:1, based on static load. The notation “LIFT HERE” or “TIE DOWN” shall be stenciled, in black letters, near each lifting and tie-down device.

3.23 Identification plate. A corrosion resistant identification plate shall be permanently installed at an easily accessible place on the forklift. The identification plate shall contain the following information:

a. Nomenclature.
b. Manufactures Name
c. Manufacturer’s Address
d. Manufacturer’s Phone Number.
e. Manufacturer’s Model Number.
f. Manufacturer’s Serial Number.
g. Registration Number.
h. National Stock Number.
i. Vehicle Curb Weight.
j. Payload, Maximum (lbs).
k. Gross Weight, Maximum (lbs).
l. Date of Delivery.
m. Warranty (months/miles).
n. Contract Number.
o. Ship to_____ Mark for _____.
p. US Property.

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3.24 Markings. All stencils, decals, plates, etc. normally provided on the manufacturer's standard forklift shall be included. This shall include, at a minimum, the forklift's capacity in pounds, and all information, cautions and warnings normally provided.

3.25 Performance. The forklift shall be capable of meeting the following performance requirements:

a. Rated load: Capability to safely handle a 2,000 pound, 48 inch per side cube with the center of gravity at the center of the cube. This shall be defined, and referred to in this document, as the rated load. The forklift shall be able to safely handle this rated load at all lift heights and fork side shift positions that it is capable of attaining.

b. Load lifting speed: Load lifting speed shall be at least 30 feet per minute with a raised load.

c. Load lowering speed: Load lowering speed shall be not more than 80 feet per minute with a rated load on the forks and not less than 20 feet per minute with no load on the forks.

d. Right angle turn: With a rated load on the forks and forklift located perpendicular to a wall, and with the front of the load against the wall, the truck shall be able to back up and make a complete right angle turn (ending up parallel to the wall) within 136 inches.

e. Travel speed: The forklift shall travel at a speed of at least 5 miles per hour, in both forward and reverse directions, while carrying a rated load on the forks.

f. Slope ascension: The forklift shall be able to accelerate up a 15 percent slope from a dead stop, with the truck facing up the slope, while carrying a rated load on the forks.

g. Upright tilt: The forks shall have at least 2 degrees forward tilt, and at least 6 degrees of rear tilt with no load on the forks.

h. Lowest upright height (collapsed mast height): The dimension from ground to top surface of the uprights, with uprights vertical, shall not exceed 86 inches with the unloaded forks fully lowered.

i. Forklift height. The forklift height shall be at least 144 inches with a rated load on the forks and the forks horizontal, when measured from ground to top surfaces of the forks.

j. Free lift height. The height that the forks can raise the rated load before the collapsed mast height is increased, shall be at least 40 inches.

k. Drift of load. With hydraulic fluid at ambient temperature, lift assembly shall hold rated load at maximum lift height for at least 10 minutes with not more than 1 to 3/4 inch of vertical drift, and not more than 1 degree of rotational drift.

l. Stability. Forklift shall meet the ANSI/ITSDF B56.1 "forward stacking", "forward travel", "lateral stacking", and "lateral travel" stability requirements.

m. Electromagnetic radiation. Forklift shall comply with the EMI requirements shown in Table I.

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TABLE I. Electromagnetic interference (EMI).

Name	Description	Parameters
SAE J551-1	Compatibility	N/A
SAE J551-2	Emissions	Frequency Range: 30 to 1000 MHz
SAE J551-11	Susceptibility	Frequency Range: (100 kHz to 18 GHz) 20 V/m (100 kHz to 200 MHz) 50 V/m (above 200 MHz)

3.26 Tires. The tires shall be pneumatic, and tire loading shall not exceed those specified in the Tire and Rim Association (TRA) Yearbook.

3.27 Cybersecurity. The forklift shall be designed, produced, and delivered with cybersecurity considerations in accordance with DoDI 8500.01, DoDI 8510.01, and AFI 17-101 and referenced security controls for applicable Information Technology (IT) and/or Operational Technology (OT), commonly referred to as Platform Information Technology (PIT). Commercial policy and standards, such as ISO/SAE 21434 and other cybersecurity related ISO/SAE/IEC standards, may be used in lieu of DoDI/AFI, as applicable.

3.27.1 Secure communication. The forklift shall be designed, produced, and delivered with the bare minimum of any wireless communication (i.e. cellular, Wi-Fi, telematics, Bluetooth, etc.) while still meeting operational/performance specifications within this CID. Any wireless communications remaining within the final configuration of the forklift shall be secured to mitigate known vulnerabilities and threats in accordance with applicable policy and standards as referenced in paragraph 3.27. All risks associated with known vulnerabilities and potential threats of any wireless communication capability shall be identified, assessed, and reported.

3.27.2 Hardware supply chain integrity. The forklift shall be designed, produced, and delivered with security controls for the supply chain integrity of electronics hardware (i.e. electronics control units (ECUs)) that involve memory storage, data processing, and/or interfaces with the communication system or controller area network (CAN) bus within the vehicle. The forklift shall be produced with mitigation to the risk of counterfeit electronic parts being installed in end items or otherwise entering the United States Air Force (USAF)/Department of Defense (DoD) inventory. The contractor shall provide products and components, or sub-components, sourced through original equipment manufacturers or authorized resellers.

3.27.3 Software/firmware supply chain integrity. The forklift shall be designed, produced, and delivered with security controls for the supply chain integrity of all delivered software and firmware. All software and firmware shall function as intended, and shall be free of vulnerabilities (i.e. defects, bugs, malware, etc.), either intentionally or unintentionally inserted as part of the software/firmware throughout the lifecycle. The contractor shall provide products and components, or sub-components, sourced through original equipment manufacturers or authorized resellers.

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4. REGULATORY REQUIREMENTS.

4.1 Recovered, recycled, or biobased materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with the Federal Acquisition Regulations (FAR) part 23.403. However, used, rebuilt or remanufactured components, pieces, and parts shall not be incorporated in the forklift.

5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product conformance. The products provide shall meet all the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards and quality assurance practices, and shall be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

5.2 Market acceptability. The following market acceptability criteria are necessary to document the quality of the product to be provided under this CID.

5.2.1 Commercial. The vehicle furnished shall be, as of the date of award, the latest model (with minor modifications, as that term is defined in DoD FAR Supplement 252.211-7012, paragraph K-85 of the solicitation) of an established product previously produced and sold in substantial commercial quantities (exceeding the criteria at FAR 15.804-3(f)(2)(ii)(A)) for at least three years. The vehicle furnished, or earlier models of it, absent minor modifications, shall have routinely been supported with spare/repair parts which were produced or sold in the normal course of business. At the time of delivery, the contractor shall furnish the Administrative Contracting Officer (ACO) with verification of compliance with these requirements. Offers shall provide, as part of their proposals, the following information on the vehicle (or its earlier models) being offered as a commercial item:

a. Total sales to the U.S. Government or to contractors for U.S. Government use during the three years.
b. Total sale of the item to the general public during the three years.
c. Length of time the item has been sold in the commercial market place.

6. PACKAGING. Preservation, packing and marking shall be specified in the contract or order.

7. NOTES.

7.1 Intended use. The forklift truck is intended for use in handling 2,000-lb capacity cargo in and around warehouses, loading platforms, docks, and on paved or other hard surfaces.

7.2 Source of documents.

7.2.1 Air Force Instructions (AFI). Electronic copies of AFIs are available online at https://www.e-publishing.af.mil/.

7.2.2 American National Standards Institute/Industrial Truck Standards Development Foundation (ANSI/ITSDF) standards. Copies are available online at https://www.itsdf.org/ or from Industrial Forklift Standards Development Foundation (ITSDF), Suite 460, 1750 K Street NW, Washington, DC 20006.

7.2.3 Electronic Code of Federal Regulations (eCFR). Electronic copies of the eCFR are available online at https://www.eCFR.gov/.

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7.2.4 Department of Defense Instructions (DoDI). Electronic copies of DoDIs are available online at https://www.esd.whs.mil/DD/.

7.2.5 Department of Defense Federal Acquisition Regulation (DFAR) and Federal Acquisition Regulation (FAR). Electronic copies of the DFAR and FAR may be obtained from https://www.acquisition.gov/.

7.2.6 Government documents. Electronic copies of specifications, standards, and handbooks are available online at https://assist.dla.mil/.

7.2.7 Occupational Safety and Health Association (OSHA). Copies of OSHA documents may be obtained from Occupational Safety and Health Association, 200 Constitution Ave., NW, Washington, DC 20560, or online at https://www.osha.gov/.

7.2.8 SAE International standards. Copies of SAE standards are available online at https://www.sae.org/ or from SAE International, 400 Commonwealth Drive, Warrendale PA 15096.

7.2.9 Tire and Rim Association (TRA) standards. Copies of TRA standards may be obtained from the Tire and Rim Association, Inc., 175 Monroe West Avenue, Suite 150, Copley, OH 44321, or online at https://www.us-tra.org/publications.html

7.2.10 Underwriters' Laboratories (UL). Copies of UL standards are available online at https://www.shopulstandards.com/ or from Comm 2000, 151 Eastern Avenue, Bensenville, IL 60106.

7.3 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this CID.
b. Final color other than yellow (see 3.20).
c. Additional battery (see 3.3).
d. Battery charger (see 3.3).
e. Packaging requirements (see 6).

7.4 National stock numbers (NSNs). The following NSNs are associated with this document. The list may not be indicative of all possible NSNs associated with this document.

NSN	DOCUMENT NUMBER	DOCUMENT STATUS
3930-00-678-2580	MIL-T-21869-35	Cancelled
3930-01-455-5463	A-A-59141	Active

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7.5 Key words.

Electric powered forklift
Rear wheel steer forklift
Pneumatic-tires warehouse forklift
2000 Pound cargo handling

7.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

MILITARY INTERESTS:

Custodians:
Army - AT
Navy - SA
Air Force - 184

Review activities:
DLA – CQ

CIVIL AGENCY COORDINATING ACTIVITY

Preparing activity:
Air Force – 184

Agent:
Air Force – 110

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